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# Swin-LLIE: Illumination-Aware Swin Transformer for Adaptive Low-Light Image Enhancement

## Abstract

Low-light image enhancement (LLIE) remains a challenging problem in computer vision due to the complex degradation of visual signals, characterized by low visibility, intensive noise, and color distortion. While Convolutional Neural Networks (CNNs) have dominated this field, they often suffer from limited receptive fields, failing to capture long-range dependencies necessary for global illumination adjustment. Conversely, standard Vision Transformers (ViTs) offer global context but incur prohibitive computational costs and often lack specific inductive biases for illumination modeling. In this paper, we propose **Swin-LLIE**, a novel architecture that synergizes the efficiency of Hierarchical Swin Transformers with the physical intuition of Retinex theory. Central to our approach is the **Illumination-Guided Attention Module (IGAM)**, a mechanism that spatially modulates transformer features based on a learnable illumination prior. By integrating IGAM into a U-shaped Swin Transformer backbone and employing a hybrid loss function comprising spatial, perceptual, and chromatic terms, Swin-LLIE achieves region-adaptive enhancement. Theoretical analysis and architectural design demonstrate that Swin-LLIE effectively balances noise suppression in dark regions with detail preservation in well-lit areas, offering a robust solution for restoration in non-uniform lighting conditions.

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## 1. Introduction

The ubiquity of digital imaging in surveillance, autonomous driving, and mobile photography has necessitated robust algorithms capable of operating in suboptimal lighting conditions. Images captured in low-light environments suffer from a low signal-to-noise ratio (SNR), resulting in loss of detail, color bias, and severe noise amplification [1].

The mathematical formulation of this problem is to recover a well-lit image  $I_{gt}$  from a degraded observation  $I_{low}$ . Traditional methods, such as Histogram Equalization (HE) and Gamma Correction, often apply global adjustments, ignoring the spatially variant nature of illumination [2]. Retinex-based methods [3] decompose images into reflectance and illumination components but often introduce halo artifacts and unnatural operational complexity.

Deep learning approaches have recently advanced the state-of-the-art. CNN-based methods [4, 5] learn direct mappings or residual corrections. However, convolution operations possess a fundamental limitation: they process information locally. In LLIE, enhancing a dark pixel effectively often requires contextual information from distant light sources or similar textures elsewhere in the image.

To address this, we introduce **Swin-LLIE**, an architecture that leverages the **Swin Transformer** [6]. Unlike standard Transformers which compute global self-attention with quadratic complexity  $O(N^2)$ , Swin Transformers utilize window-based attention with linear complexity  $O(N)$ , making them suitable for high-resolution image restoration.

### Our primary contributions are:

1. **Illumination-Guided Attention Module (IGAM)**: A novel feature modulation unit that injects physics-based illumination priors into the deep feature space, allowing the network to distinguish between dark regions (requiring aggression enhancement) and

bright regions (requiring preservation).

2. **Hybrid Architecture:** We propose a U-Net structured Swin Transformer that integrates IGAM at multiple scales, utilizing a specific Cross-Stage Feature Fusion (CSFF) to mitigate information loss during downsampling.
  3. **Refined Loss Landscape:** We employ a comprehensive objective function incorporating L1 reconstruction, VGG perceptual loss, and a cosine-similarity-based color consistency loss to ensure structural and chromatic fidelity.
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## 2. Related Work

### 2.1 Traditional and CNN-based Enhancement

Early works relied on statistical models. HE and Contrast Limited Adaptive Histogram Equalization (CLAHE) stretch dynamic range but amplify noise. Retinex theory [3] assumes an image  $I = R \circ L$  and attempts to estimate  $L$  to recover  $R$ . Deep RetinexNet [4] unfolded this optimization into a neural network. Zero-DCE [7] formulated enhancement as the estimation of pixel-wise curve adjustments. While efficient, these CNN methods often lack the capacity to model long-range dependencies due to the fixed receptive field of convolution kernels.

### 2.2 Vision Transformers

The Vision Transformer (ViT) [8] introduced self-attention to computer vision, enabling global receptive fields. However, its quadratic complexity limits application in dense prediction tasks. The Swin Transformer [6] addressed this via shifted windowing mechanisms. SwinIR [9] demonstrated the efficacy of Swin Transformers in super-resolution and denoising. Swin-LLIE extends this lineage by introducing task-specific inductive biases—specifically, illumination awareness—into the transformer block, addressing the unique heterogeneity of low-light scenes.

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## 3. Methodology

### 3.1 Problem Formulation and Architecture Overview

We define the enhancement task as learning a residual mapping function  $\mathcal{F}_\theta$ . Given a low-light input  $I_{low} \in \mathbb{R}^{H \times W \times 3}$ , the network predicts a residual  $\mathbf{I}$  such that the enhanced image  $\hat{I} = I_{low} + \mathbf{I}$ . The residual learning formulation is chosen to stabilize training, as the network need only learn the high-frequency difference and illumination boost rather than synthesizing the full image manifold.

The architecture (Fig. 1) follows a hierarchical U-Net design comprising three main stages:

1. **Illumination Estimation Module (IEM):** Generates a spatial prior.
2. **Swin-U-Net Encoder-Decoder:** The core feature extraction backbone.
3. **IGAM Integration:** Modulates features based on the IEM output.

### 3.2 Illumination Estimation Module (IEM)

To make the network “illumination-aware,” we must first approximate the illumination distribution  $L$ . We employ a dual-branch strategy combining physical priors with learned refinement.

**Branch 1 (Retinex Prior):** We utilize the Max-RGB assumption, which posits that the maximum value across color channels approximates the intensity of the incident light:

$$L_{rough}(x) = \max_{c \in \{R, G, B\}} I_c(x)$$

**Branch 2 (Learned Refinement):** A shallow CNN processes the input to correct the Max-RGB estimate, utilizing dilated convolutions to expand the receptive field and ensure

local smoothness.

The final illumination map  $L$  is inverted to create a **Dark Mask**  $M$ , where high values indicate regions requiring enhancement:

$$M = 1 - \text{sigmoid}(\text{Conv}_{\text{refine}}(I_{\text{low}}) + L_{\text{rough}})$$

### 3.3 Swin Transformer Backbone

The backbone utilizes Swin Transformer blocks to process features. The core innovation of Swin is replacing global Multi-Head Self-Attention (MSA) with Window-based MSA (W-MSA) and Shifted Window MSA (SW-MSA).

Given an input feature map, it is partitioned into non-overlapping windows of size  $M \times M$  (typically  $M = 8$ ). The attention within a window is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) + \text{Bias} V$$

where  $Q, K, V$  are query, key, and value matrices, and  $B$  is the learnable relative position bias.

To enable cross-window information exchange, alternating blocks shift the window partitioning by  $(M/2, M/2)$ . This shifted window mechanism reduces computational complexity from  $O(H^2W^2)$  to  $O(HW)$ , linear with respect to image resolution, while maintaining global connectivity over consecutive layers.

### 3.4 Illumination-Guided Attention Module (IGAM)

The IGAM is the theoretical core of Swin-LLIE. Standard transformers treat all patches equally (content-adaptive but not explicitly illumination-adaptive). IGAM explicitly re-weights features based on the Dark Mask  $M$ .

Let  $F_{in} \in \mathbb{R}^{C \times H \times W}$  be the intermediate feature map. IGAM performs a channel-wise and spatial modulation:

1. **Channel Descriptor:** We compute channel-wise statistics  $\alpha \in \mathbb{R}^C$  using a global average pooling followed by an MLP (Dense-ReLU-Dense-Sigmoid), similar to Squeeze-and-Excitation networks. This determines *which* channels contain relevant enhancement information.
2. **Spatial Refinement:** The global Dark Mask  $M$  is adapted to the current feature resolution via a convolution layer, yielding  $M'$ .
3. **Modulation:**

$$F_{out} = F_{in} \odot (1 + \alpha M')$$

Here,  $\odot$  denotes element-wise multiplication. The term  $(1 + \dots)$  ensures that in well-lit regions (where  $M \approx 0$ ), the features pass through identity ( $F_{out} \approx F_{in}$ ), preserving original details. In dark regions (where  $M \approx 1$ ), features are amplified by a factor of  $(1 + \alpha)$ . This multiplicative modulation preserves feature statistics better than additive bias.

### 3.5 Cross-Stage Feature Fusion (CSFF)

In U-Net architectures, skip connections concatenate encoder features ( $F_{enc}$ ) with decoder features ( $F_{dec}$ ). However,  $F_{enc}$  contains noisy low-level information, while  $F_{dec}$  contains reconstructed high-level semantics. Simple concatenation is suboptimal.

CSFF introduces a learned gating mechanism  $g \in [0, 1]$ :

$$F_{fused} = g \odot F_{enc} + (1 - g) \odot F_{dec} + \text{Conv}(F_{enc} \parallel F_{dec})$$

This allows the network to selectively suppress noise from the encoder features before fusing them with the decoder.

## 4. Loss Function Design

Training is driven by a hybrid loss function  $\mathbf{l}_{total}$  designed to constrain the solution space across multiple domains:

$$\mathbf{l}_{total} = \lambda_1 \mathbf{l}_{L1} + \lambda_2 \mathbf{l}_{percep} + \lambda_3 \mathbf{l}_{color} + \lambda_4 \mathbf{l}_{tv}$$

1. **Reconstruction Loss ( $\mathbf{l}_{L1}$ ):**

$$\mathbf{l}_{L1} = \| \hat{I} - I_{gt} \|_1$$

The L1 norm is preferred over L2 as it induces sparsity and prevents the blurring effects associated with regression-to-mean.

2. **Perceptual Loss ( $\mathbf{l}_{percep}$ ):** Defined over the feature maps  $\phi_j$  of a pre-trained VGG-19 network at layers  $j \in \{3, 8, 15, 22\}$ :

$$\mathbf{l}_{percep} = \sum_j \| \phi_j(\hat{I}) - \phi_j(I_{gt}) \|_2^2$$

This ensures the enhanced image matches the ground truth in semantic feature space, preserving textures and edges.

3. **Color Consistency Loss ( $\mathbf{l}_{color}$ ):** To correct color shifts (e.g., green/purple haze in dark regions), we enforce cosine similarity between RGB vectors of the output and ground truth pixels:

$$\mathbf{l}_{color} = \sum_p \left( 1 - \frac{\hat{I}_p \cdot I_{gt,p}}{\|\hat{I}_p\|_2 \|I_{gt,p}\|_2 + \epsilon} \right)$$

This metric is intensity-invariant, focusing purely on chromatic alignment.

4. **Illumination Smoothness Loss ( $\mathbf{l}_{tv}$ ):** Total Variation (TV) loss is applied to the estimated illumination map  $L$  to enforce piecewise smoothness and suppress noise in the prediction:

$$\mathbf{l}_{tv} = \| \nabla_x L \|_1 + \| \nabla_y L \|_1$$

## 5. Implementation Details

**Model Specifications:** The model uses patch size  $4 \times 4$ , with an embedding dimension of  $C = 60$ . The encoder depths are [4,4,4] Swin blocks, and the attention head counts are [6,6,6]. The window size is set to  $M = 8$ . The total parameter count is approximately 4.8 million.

**Training Protocol:** \* **Optimizer:** AdamW with  $\beta_1 = 0.9, \beta_2 = 0.999$ . \* **Learning Rate:** Initial  $2 \times 10^{-4}$  with Cosine Annealing decay. \* **Data Augmentation:** Random rotation, flipping, and cropping to  $96 \times 96$  patches. \* **Platform:** PyTorch framework on NVIDIA RTX 3090 GPU.

## 6. Theoretical Analysis and Discussion

### 6.1 Complexity vs. Performance

Standard ViTs scale quadratically. For a 1080p image, a global attention map is computationally intractable. Swin-LLIE’s linear complexity  $O(HW)$  allows it to process high-resolution inputs typical of modern photography. Furthermore, the incorporation of IGAM adds negligible cost ( $O(C)$ ) while providing a significant performance boost by acting as a “soft attention” mask that guides the heavy transformer computation toward relevant spatial regions.

### 6.2 Ablation Implications

We posit that removing IGAM would result in a “flat” enhancement, where dark regions remain under-exposed or bright regions become saturated. Similarly, removing the CSFF

would likely result in the loss of high-frequency texture details, as the decoder would rely solely on upsampled features without the residual high-frequency information from the encoder.

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## 7. Conclusion

In this work, we presented **Swin-LLIE**, a comprehensive framework for low-light image enhancement. By bridging the gap between physical illumination models (Retinex) and modern representation learning (Swin Transformers), we achieve a balance of restoration quality and computational efficiency. The proposed IGAM module serves as an effective mechanism for spatially adaptive processing, ensuring that enhancement is applied only where needed. Swin-LLIE represents a significant step forward in robust, physics-aware deep learning for image restoration.

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